Kalyan

Slides 9 and 10 · then narrate the demo while Diya clicks

Slide 9 — Database design decisions

40 sec

- **Why SQL:** a cart line is meaningless without its cart and product. Foreign keys, UNIQUE and CHECK do the work for us.
- **Why SQLite:** one file, no setup. Every test makes its own copy.
- **Say the trade-off yourself:** "SQLite handles one writer at a time. Fine for one store. A bigger deployment moves to MySQL and the schema ports straight over."

Slide 10 — ERD

60 sec · worth 5 marks

- "**Eight entities.** Three for the shop, two for orders, three for accounts."
- "`cart_items` is the junction table — it joins carts to products."
- "`order_items` copies the product name and price at checkout, so the order keeps what the customer actually paid."

That last line is the one that impresses. Do not skip it.

Slide 19 — Your demo lines

Diya clicks, you talk

1	"Eight products, read from SQLite."
2	"The cart now belongs to this account."
3	"Two new rows in `cart_items`."
4	"The total came back recalculated from the API."
5	"The row is gone. Not hidden — deleted."
6	"Stock is three. The server refused the fourth."
7	**"One click, three tables. All in one transaction."** ← slow down here
8	"The same order, read back from the API."

Your questions

Q4	How many entities? — **Eight.**
Q5	Primary key? — Uniquely identifies a row. `product_id`.
Q6	Foreign key? — Points at another table's primary key. `cart_items.cart_id` → `carts.cart_id`.
Q7	One relationship? — One cart has many cart_items. Each line belongs to one cart.
Q8	How does the database help? — It refuses bad data itself, even if the API has a bug.
Q12	Why this stack? — Relational data needs SQL. SQLite needs no setup. Node means one language for the whole project.

Q21 / Q22 are personal. Yours: Week 2 data analysis, GST in the cart API, Sprint 4 integration (PR #46, #49, #53).